

AS Chemistry homework · 3.5 Shapes of molecules

CIE 9701 AS Chemistry · 18 marks · about 40 minutes

Topic: 3.5 Shapes of molecules — VSEPR theory, bond pairs, lone pairs, electron-pair geometry and molecular shape (including BF₃, CO₂, CH₄, NH₃, H₂O, SF₆, PF₅ and analogues). Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; use VSEPR language throughout.

Section	Content	Marks
A	Multiple choice	8
B1	2021 ON Paper 21 Question 1(a)(iii)	1
B2	2024 MJ Paper 21 Question 2(b)(i)	2
B3	2024 MJ Paper 21 Question 2(b)(ii)	2
B4	2023 MJ Paper 21 Question 1(e)	2
B5	2025 FM Paper 22 Question 1(b)(ii)	1
B6	2025 MJ Paper 23 Question 4(b)(i)	2
Total		18

Section A — Multiple-choice questions (8 marks)

A1.

[1]

7 Which row is correct?

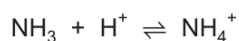
	smallest bond angle	→	largest bond angle
A	H ₂ O	NH ₃	BF ₃
B	BF ₃	NH ₃	H ₂ O
C	NH ₃	H ₂ O	BF ₃
D	H ₂ O	BF ₃	NH ₃

2025 MJ Paper 12 Question 7

A2.

[1]

5 Ammonia reacts with acids to form the ammonium ion.



Which row is correct?

	shape of NH ₄ ⁺	bond angle in NH ₄ ⁺ / °
A	pyramidal	107
B	pyramidal	109.5
C	tetrahedral	107
D	tetrahedral	109.5

2023 MJ Paper 13 Question 5

A3.

[1]

24 What is the bond angle in the ammonium ion?

A 90° **B** 107° **C** 109.5° **D** 120°

2024 MJ Paper 11 Question 24

A4.

[1]

11 Which statement about the molecule PF_5 is correct?

- A** Every F–P–F bond angle is 90° .
B The central atom in the molecule does **not** have a lone pair of electrons.
C The molecule has an overall dipole moment.
D The shape of the molecule is octahedral.

2025 MJ Paper 14 Question 11

A5.

[1]

34 Which molecules contain at least one bond angle of 120° ?

- 1 C_2H_4
2 PF_5
3 NCl_3

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

Use of the Data Booklet may be appropriate for some questions.

2021 MJ Paper 11 Question 34

A6.

[1]

3 In which pair is the bond angle in the first species smaller than the smallest bond angle in the second species?

- A** CH_4 and SF_6
B CO_2 and BF_3
C H_2O and H_3O^+
D NH_4^+ and NH_3

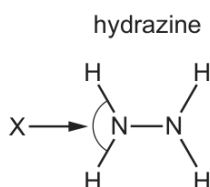
2025 MJ Paper 14 Question 3

A7.

[1]

6 VSEPR theory should be used to answer this question.

Hydrazine has the following structure.



What is the predicted bond angle X?

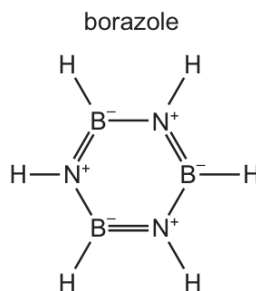
- A** 90° **B** 107° **C** 109.5° **D** 120°

2022 MJ Paper 13 Question 6

A8.

[1]

- 6 A structure for borazole, $\text{N}_3\text{B}_3\text{H}_6$, is shown.



Which shape is borazole and how many π electrons are there in the structure?

	shape	number of π electrons
A	non-planar	3
B	non-planar	6
C	planar	3
D	planar	6

2024 MJ Paper 13 Question 6

Section B — Structured questions (10 marks)

B1(a)(iii).

[1]

- 1 Sulfides are compounds that contain sulfur but not oxygen.

(a) Carbon disulfide, CS_2 , is a volatile liquid at room temperature and pressure.

(iii) Suggest the bond angle in a molecule of CS_2 .

..... [1]

2021 ON Paper 21 Question 1(a)(iii)

B2(b)(i).

[2]

(b) A molecule of SCl_2 contains two S–Cl covalent bonds.

- (i) Complete the dot-and-cross diagram in Fig. 2.1 to show the arrangement of the outer electrons in a molecule of SCl_2 .

Use $\times$ to show electrons from the chlorine atoms.

Use $\bullet$ to show electrons from the sulfur atom.

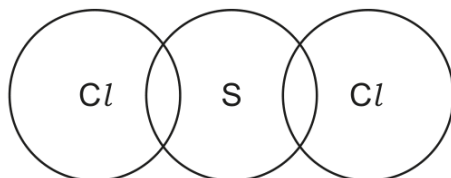


Fig. 2.1

[2]

2024 MJ Paper 21 Question 2(b)(i)

B2(b)(ii).

[2]

(b) A molecule of SCl_2 contains two S–Cl covalent bonds.**(ii)** Predict the shape of, and bond angle in, a molecule of SCl_2 by using VSEPR theory.

shape

bond angle

[2]

2024 MJ Paper 21 Question 2(b)(ii)

B4(e).

[2]

- 1** Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(e) Te reacts with F_2 at 150°C to form TeF_x . Molecules of TeF_x are octahedral with bond angles of 90° .Explain why TeF_x is octahedral with bond angles of 90° .

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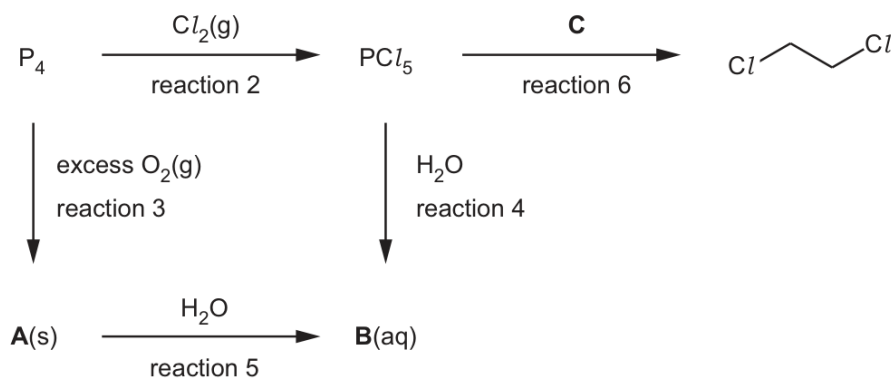
..... [2]

2023 MJ Paper 21 Question 1(e)

B5(b)(ii).

[1]

- 1** Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .**Fig. 1.1****(ii)** Predict the shape of a molecule of PCl_5 .

..... [1]

2025 FM Paper 22 Question 1(b)(ii)

B6(b)(i).

[2]

- 4 The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.



Fig. 4.1 shows how the concentration of $\text{HCl}(\text{aq})$ varies with time.

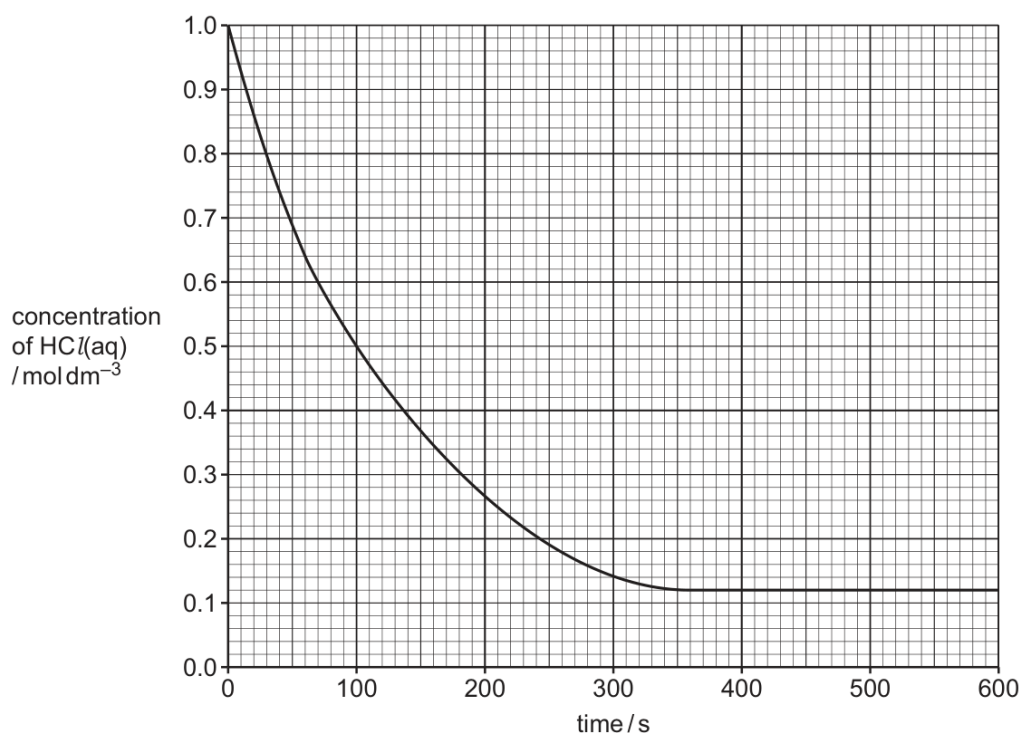


Fig. 4.1

- (b) Fig. 4.2 shows a possible arrangement of outer-shell electrons in one SO_2 molecule.

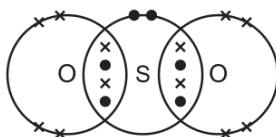


Fig. 4.2

- (i) Use Fig. 4.2 to predict the shape and bond angle of a molecule of SO_2 .

shape

bond angle°

[2]